



Bioinformatics Script - Designing Safe CDS: Logic, Data, and the Consequences of Error

The following is a script of the bioinformatics project, and all information in this document is subject to change.

Designing Safe CDS: Logic, Data, and the Consequences of Error

Total Duration: ~10 minutes

Characters:

- **Player** - Silent, acting as a Student Nurse.
- **Nurse Sarah** - Charge Nurse; professional, fatigued by technology.
- **Mr. Johnson** - 68-year-old male patient, lying in bed.
- **Narrator** - Disembodied voice. Also narrates relative EHR alerts.

Actions: Throughout the script, gameplay elements that require the player's action have been underlined.

EHR Terminal State Color Key:

- EHR Locked State
 - Pink - Paused
 - Red Alert
 - Orange - False Positive
 - Red - High Severity
- Yellow - Scanner State
- Purple - Notes Review/Correct Dosage State
- Blue - Override State

SCENE 1 - INTRO & THE FLAWED ALERT (Methotrexate)

Setting

A realistic hospital room, morning. Sunlight through blinds.

Visuals: Mr. Johnson resting in bed (idle animation). A Computer on Wheels (COW) with an EHR terminal sits bedside. A medication cart is nearby. Nurse and patient have a static name tag that turns toward the player.

Audio: Low AC hum, distant paging, soft monitor beeping.

[FADE IN]

NARRATOR

Welcome to the “Designing Safe Clinical Decision Support” VR experience. Clinical Decision Support, or CDS, refers to software built into the Electronic Health Record that provides clinicians with patient-specific alerts, reminders, and recommendations. When designed well, CDS helps prevent medication errors and improves patient safety. When designed poorly, it can introduce new risks.

NARRATOR

In this world, you are a student nurse shadowing the charge nurse, Sarah. This is your first ever shift. In order to interact with objects, wait for them to glow, then point and click either of your trigger buttons.

NARRATOR

We are about to begin. To pause at any time, click _____. Enjoy!

[Diagram of button is shown during above narration. Nurse Sarah walks into the room and logs into the computer while talking. She turns to the Player.]

NURSE SARAH

Glad you could join me. I’m Sarah and our patient this morning is Mr. Johnson. He’s stable, but let’s get his morning meds sorted.

[She gestures toward the Med Cart and EHR screen.]

NURSE SARAH

First, we scan the wristband, then scan the meds, and the system *should* catch any problems. Lately, it’s been more of a headache than help... you’ll see.

[She looks directly at the Player.]

NURSE SARAH

Go ahead. Pick up the scanner.

NARRATOR

Click on the scanner using the trigger button.

[Scanner glows/pulses yellow. Wait for pick up input.]

NURSE SARAH

Now, you can scan Mr. Johnson's wristband.

[Wristband glows/pulses yellow. The player scans wristband.]

[SFX: Beep!]

[EHR: "PATIENT VERIFIED: JOHNSON, M. (Male, 68)"]

NURSE SARAH

Good. Now scan the Methotrexate.

[Methotrexate bottle glows/pulses. Wait for scanning input.]

[SFX: Beep! A red triangle symbol appears on the computer and pulses.]

[A large RED ALERT pops up in 3D space above the computer/EHR terminal: "WARNING: Patient is of childbearing age. Methotrexate is teratogenic." pops up on the computer screen.]

[Sarah sighs, shakes her head.]

NURSE SARAH

See? "Childbearing age"? He's a 68-year-old man.

[She dismisses the alert by interacting with the keyboard]

NURSE SARAH

Makes no sense. The system doesn't even check gender. Just override it while I give the medication.

[Nurse Sarah administers Methotrexate to the patient while the keyboard glows. Player overrides the alert by clicking on the keyboard. Click sound plays.]

[Nurse Sarah returns to the computer.]

[SIMULATION PAUSE - QUIZ]

Question: *What caused this faulty alert?*

A. Patient is allergic

Narrator upon answering: Incorrect. There is no documented allergy for this patient. The alert was triggered before any allergy logic was applied.

B. Logic failed to filter for gender

Narrator upon answering: Correct. The CDS logic checked age but failed to account for patient sex, triggering a teratogenic warning for a male patient. This type of poorly scoped logic creates false alerts and undermines trust in the system.

C. Medication expired

Narrator upon answering: Incorrect. Medication expiration would generate a different alert. This warning was unrelated to the medication's age or storage.

[Player selects answer, and narrator says the corresponding prompt.]

[SIMULATION RESUMES]

[Alert disappears.]

NURSE SARAH

The trick is knowing when to trust the system... and when to trust your judgment. Let's keep going.

SCENE 2 - THE MANUAL ENTRY TRAP (Amoxicillin)

NURSE SARAH

Next up: Amoxicillin. Go ahead and grab the bottle.

[Bottle glows/pulses yellow. The player scans it.]

[SFX: Beep!]

[EHR: "Prescription: Amoxicillin, Notes (free text): 'Last dose: 5000mg', Input: A numeric keypad appears."]

NURSE SARAH

Check the history notes for his last dose, then enter the current one.

[Player sees "5000mg" and enters "5000."]

NURSE SARAH

Whoa - stop! That's a huge dosage. That "5000mg" is a typo in free text. It should be 500mg. But because it isn't structured data, the system didn't flag it. A 5000mg dose would destroy his kidneys.

NURSE SARAH

Correct it to 500mg.

[Keyboard glows]

[Player corrects dose by clicking the keyboard and confirms. A click sound is played.]

[SFX: Pleasant chime.]

[EHR: "Dosage Confirmed."]

NURSE SARAH

Better. But the system shouldn't have let you type that in without screaming at you.

SCENE 3A - CONTRAINDICATION & ALERT FATIGUE (Allopurinol)

NURSE SARAH

Last one. Scan the Allopurinol.

[Bottle glows/pulses. Player scans.]

[SFX: Big beep!]

[EHR: "WARNING!! CONTRAINDICATION! Amoxicillin is contraindicated with Allopurinol. Do not administer."]

[Sarah looks exhausted and waves it off.]

NURSE SARAH

The software's been wrong all morning. It's probably another bad flag.

[Keyboard glows/pulses red. Player presses it, and Nurse Sarah goes on to administer the medications.]

NURSE SARAH

Okay, meds administered. Let's move on.

[FADE TO BLACK]

[Text: "30 Minutes Later..."]

SCENE 3B - CONSEQUENCE

[FADE IN - Emergency ambience]

Audio: Fast alarm tones, labored breathing.

Visuals: Room dimmer, red tint. Mr. Johnson covered in rashes.

[Sarah stands near the monitor, panicked.]

NURSE SARAH

His blood pressure is dropping! This rash - could be anaphylaxis!

[She scrolls through the EHR, horrified.]

NURSE SARAH (whispering)

Oh no... the alert. It wasn't wrong. It was a real contraindication!

[She turns to the Player.]

NURSE SARAH

We ignored it because of the others... and we hurt him.

[She starts crying, mumbling "I'm gonna lose my job, how could I do this?" etc.]

NARRATOR

Poor Mr. Johnson... :(

[FADE OUT SLOWLY]

SCENE 4 - POST-EXPERIENCE ASSESSMENT

Neutral void or quiet office space.

UI - EXPERIENCE COMPLETE

Select your Major:

- Nursing
- Healthcare Informatics / Computing

[Different questions appear depending on major.]

QUESTION 1: IN-SIMULATION QUIZ (METHOTREXATE ALERT)

- **Question:** What caused this faulty alert?
- **A.** Patient is allergic (Narrator: Incorrect. There is no documented allergy; the alert was triggered by medication logic, not allergy files.)
- **B.** Logic failed to filter for patient sex (Narrator: Correct. The CDS logic checked age but failed to account for patient sex, triggering a teratogenic warning for a male patient. This poorly scoped logic creates false alerts and leads to alarm fatigue.)
- **C.** Medication expired (Narrator: Incorrect. Expiration logic is separate from clinical guideline alerts.)

QUESTION 2: NURSING MAJOR PATH (ALERT FATIGUE)

- **Question:** Sarah exhibited alert fatigue. How did the false alert in Scene 1 contribute to the error in Scene 3?
- **A.** It increased Sarah's workload, making her rush.
- **B.** It caused Sarah to distrust alerts, leading her to override a valid safety check (Narrator: Correct. Repeated false alerts condition clinicians to ignore warnings reflexively. When the system fails to provide relevant info, trust is lost and safety is compromised.)
- **C.** It delayed medication administration.
- **D.** It changed the patient's allergy status.

QUESTION 3: INFORMATICS MAJOR PATH (DATA STRUCTURE)

- **Question:** The Amoxicillin error was caused by trusting unstructured free text. How should the EHR be redesigned to prevent a 5000 mg overdose entry?
- **A.** Implement spellcheck on all nursing notes.
- **B.** Use structured data fields with dose-range validation and hard stops (Narrator: Correct. Structured data must take priority. By implementing validation against the /MedicationRequest resource, the system can block entries that exceed safe limits.)
- **C.** Remove historical nurse notes to reduce clutter.
- **D.** Increase the number of manual override permissions for clinicians.

FINAL SUMMARY SCREEN

Thanks for playing! Core concepts learned:

- **Faulty CDS Logic:** Misapplied rules generate false alerts (Scene 1).
- **Manual Entry Risks:** Free-text input introduces dangerous variation (Scene 2).
- **Alert Fatigue:** Too many alerts cause clinicians to ignore real ones (Scene 3).
- **Garbage In, Garbage Out:** Poor logic and poor data = unsafe care.

[Exit simulation button appears. Upon clicking, it brings you to the main menu.]